BAMBARE

TRAVEL MANAGEMENT SYSTEM

PROJECT AGENDA

01

Project Scope

Client background, problem statement & objectives

02

Requirements

Functional & Non-Functional Requirements

03

Technologies

Tech stack, tools, languages & techniques

04

Designs & Diagrams

Architecture, ER diagrams & UI concepts

05

Project Plan

Development phases, schedule & team roles

Project Scope Client Background

10+
Years of Experience

60%
Target Booking Time Reduction

40%
Target Online Booking Increase

99.5%
Required System Uptime

CLIENT PROFILE

Company:	Bambare Travels
Client:	Mr. Dammika Bambaradeniya
Established	2015
Location:	Sri Lanka
Industry	Tourism & Travel
Speciality:	Domestic & International Tours

Project Scope Problem & Solution

⚠ Current Challenges

- Manual booking via phone/email — errors & delays
- Customer data scattered across spreadsheets
- Manual resource allocation — scheduling conflicts
- Time-consuming invoice & payment processes
- No business intelligence or analytics
- System overwhelmed during peak seasons

✓ Proposed Solutions

- Automated booking with instant confirmation
- Centralized customer data & CRM features
- Real-time resource availability & assignment
- Automated invoice generation & payment tracking
- Analytics dashboard & business intelligence
- Scalable cloud architecture for peak demand

Functional Requirements

FR1:

User Management & Authentication

- User registration with email verification
- Secure login — Admin / Customer roles
- User profile CRUD operations
- Password reset & account recovery

FR2:

Tour & Activity Planning & Booking

- Browse, search & filter tour packages
- Real-time availability & calendar check
- Online booking with instant confirmation
- Booking history & cancellation workflow

FR3:

Map, Navigation & Places Recommendation

- Google Maps API — location & navigation
- Interactive map with location pinning
- Recommended places CRUD operations
- Location-based destination discovery

FR4:

Expense Tracking & Currency Converter

- Expense tracker with 6 categories
- Real-time currency conversion via API
- Monthly expense reports & summaries
- Budget planning & tracking tools

FR5:

Admin Dashboard & Reporting

- Real-time KPI analytics dashboard
- Hotels, tours & user management CRUD
- FCM push + Email/SMS notifications
- System-wide booking & revenue reports

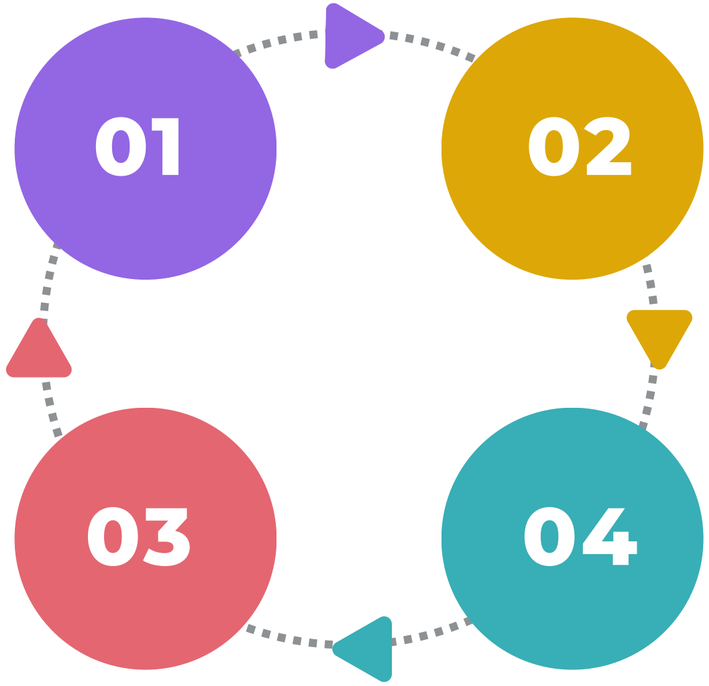
Non-Functional Requirements

Performance

- Page load < 3 seconds
- 100+ concurrent users
- DB queries < 2 seconds (95%)

Reliability

- 99.5% system uptime
- Daily automated backups
- 24-hr max recovery time



Security

- HTTPS/SSL encryption
- PCI DSS compliance
- Password hashing (industry-standard)

Usability

- Minimal training required
- Responsive on all devices
- WCAG 2.1 Level AA accessibility

Technologies, Techniques & Languages

01

Frontend

Flutter | Dart | Flutter Navigation

02

Authentication Layer

Firebase Authentication | JWT Tokens | RBAC

03

Backend Services

Node.js | Express.js RESTful APIs

04

Database Layer

Firebase Firestore (real-time) | MongoDB Atlas

05

Cloud & Integrations

Google Cloud Platform | Google Maps API | Currency API | Stripe/PayPal | FCM | SMS/Email

Technologies, Techniques & Languages

Languages

Dart

OS Target

Android 8.0+ / iOS 12.0+

IDE

VS Code

Version Control

Git & GitHub

UI Design

Figma (prototypes)

Methodology

Scrum / Agile

Cloud

Google Cloud Platform

Testing

Flutter Test, Integration Test, Postman

CI/CD

GitHub Actions

Designs &Diagrams

01

System Architecture

- AS-IS: Manual operations — phone calls, emails, spreadsheets
- TO-BE: Mobile-first cross-platform digital platform
- 5 layers: Mobile Client → Auth → Backend → Database → External APIs
- Cloud infrastructure: Google Cloud Platform with auto-scaling
- Architecture style: RESTful microservices with Firebase real-time sync

02

Use Case Diagram

- 2 primary actors: Customer & Administrator
- UC-01: User Registration & Authentication (FR1)
- UC-02: Tour Package Browsing & Booking (FR2)
- UC-03: Map Navigation & Places Recommendation (FR3)
- UC-04: Expense Tracking & Currency Conversion (FR4)
- UC-05: Admin Dashboard & Reporting (FR5)

03

ER Diagram

- 11 core entities — 3NF normalized database design
- USER ↔ CUSTOMER (1:1) | BOOKING ↔ PAYMENT (1:1)
- CUSTOMER ↔ BOOKING (1:N) | TOUR ↔ BOOKING (1:N)
- VEHICLE & GUIDE via RESOURCE_ALLOCATION junction (M:N)
- Dual database: Firebase Firestore + MongoDB Atlas

04

UI Wireframes (Figma)

- Customer registration & secure login screen
- Invoice & booking data filling interface
- Tour listing with advanced search & filter controls
- Interactive map view with location pinning
- Admin dashboard with real-time KPI widgets & charts

Project Development Plan

1 Foundation

- Project setup & repository initialization
- Firebase project & Auth configuration
- User auth module — FR1 (login, register, RBAC)
- Base UI components, theme & navigation
- ER diagram design & database schema setup
- SRS documentation v1.0 finalization
- Initial Firestore security rules configuration

3 Advanced Features

- Expense tracker & currency converter — FR4
- Admin dashboard & reporting module — FR5
- FCM push notification integration
- Automated email & SMS notification system
- System-wide integration testing
- UI/UX refinement from client feedback
- Performance profiling & optimization

2 Core Modules

- Tour browsing & booking module — FR2
- Map & navigation module — FR3 (Google Maps SDK)
- Node.js backend API development (Express.js)
- Firestore CRUD integration for tours & bookings
- Unit testing with Jest framework
- Code review sessions & sprint retrospective
- CI/CD pipeline setup via GitHub Actions

4 QA & Delivery

- User Acceptance Testing with Mr. Dammika
- Performance & load testing (500 concurrent users)
- Security audit & penetration testing
- Bug fixes & final UI polish
- Deployment to Google Cloud Platform
- API documentation (Swagger/OpenAPI)
- Final project documentation & handove

Team Roles & Function Distribution

Avishka Bambaradeniya
SA24100931

01

Map, Navigation & Places
Role: Project Manager

Prarthana Meegahakumbura
SA24101440

02

Expense Tracking & Currency
Role: UI/UX Designer

Shamla Noufer
SA24610780

03

Tour & Activity Booking
Role: UI/UX Designer

Oshidi Yapa
SA24610566

04

Admin Panel & Reporting
Role: Business Analyst

P. Vidhuja
SA24101945

05

User Management & Authentication
Role: Business Analyst

Thank You

Travel is the only thing you buy that makes you richer